

1 Basic Knowledge

Group $(G, *)$: **G1**: Closure **G2**: Associativity **G3**: Identity element **G4**: Inverse element $\downarrow$ **Division ring**: $R^\times = R \setminus \{0\}$; $R^\times$ group (under $\cdot$)

Ring $(R, +, \cdot)$: **R1**: $(R, +)$ is an abelian group **R2**: $(R, \cdot)$ is monoid. (no inverse) **R3**: Distributive law (left and right)

Ideal (I) : **I1**: $\neq \emptyset$ **I2**: $\forall a, b \in I, a - b \in I$ **I3**: $\forall r \in R, i \in I, ri, ir \in I$ **Lemma**: $I = R \Leftrightarrow 1 \in I \Leftrightarrow I \cap R^\times \neq \emptyset$

Subgroup (H) : **H1**: $e \in H$ **H2**: $\forall a, b \in H, ab \in H$ **H3**: $\forall a \in H, a^{-1} \in H$ **Subring** (S) : **S1**: $1 \in S$ **S2**: $\forall a, b \in S, a - b \in S$ **S3**: $\forall a, b \in S, ab \in S$

Integral Domain (ID): **ID1**: commutative ring **ID2**: no zero divisors **Lemma**: Finite ID $\Rightarrow$ field ID $\Rightarrow$ Cancellation Law

Factor Ring (R/I) : $a + I = b + I \Leftrightarrow a - b \in I$ $(R/I, \oplus, \odot)$ is a ring. $(1_{R/I} = 1_R + I, 0_{R/I} = 0_R + I, I \text{ ideal})$ $\mathbb{Z}_n \cong \mathbb{Z}/n\mathbb{Z}$

Ring Homomorphism: $\varphi: R \rightarrow S$ s.t. **RH1**: $\varphi(a + b) = \varphi(a) + \varphi(b)$ **RH2**: $\varphi(ab) = \varphi(a)\varphi(b)$ **RH3**: $\varphi(1_R) = 1_S$

1: $\text{Ker}(\varphi)$ is an ideal. **2**: $\text{Im}(\varphi)$ is a subring. **3**: φ is 1-1 $\Leftrightarrow \text{Ker}(\varphi) = \{0_R\}$. **FIT**: $R/\text{Ker}(\varphi) \cong \text{Im}(\varphi)$ (First Isomorphism Theorem)

Canonical Homomorphism: $\text{can}: R \rightarrow R/I$ s.t. $\text{can}(r) = r + I$ is a onto ring Homomorphism. $\text{Ker}(\text{can}) = I$ $\updownarrow$ ($\exists!$ φ 双射)

Lemma: If $\varphi: R \rightarrow S$ ring Homomorphism s.t. $I \subseteq \text{Ker}(\varphi)$, then $\exists! \tilde{\varphi}: R/I \rightarrow S$ s.t. $\varphi = \tilde{\varphi} \circ \text{can}$ (i.e. $\tilde{\varphi}(r + I) = \varphi(r)$)

Lagrange's Theorem: G finite group, $H \leq G$ subgroup, then $|H| \mid |G|$

2 Types of Ideals (Maximal Ideals $\Rightarrow$ Prime Ideals $\Rightarrow$ Radical Ideals $\Rightarrow$ Ideals)

Division: If R ring, then: $\forall f, g \in R[x], g$ 首相系数可逆 $\Rightarrow \exists q, r \in R[x]$ s.t. $f(x) = g(x)q(x) + r(x)$ and $\deg(r) < \deg(g)$ or $r = 0$.

Lemma: If R ring, $g \in R[x], g \neq 0, I = (g)$. $\Rightarrow$ $^1 \forall i \in R[x]/I, i = f + I, \deg(f) < \deg(g)$

g 首相系数可逆 2 If $a, b \in R[x], \deg(a, b) < \deg(g)$ and $a + I = b + I$, then $a = b$. (系数也相同)

Remark: If $I = (g), g$ 首相系数不可逆: Find a polynomial g_1 s.t. $g_1 = hg, h \in R[x]$ 首相系数可逆, 那么可满足 1 条件, 2 只能 subset 但不能唯一表达.

Remark: 有些时候我们能用 gcd, lcm 计算 $\langle a, b \rangle = \langle \text{gcd}(a, b) \rangle$ 和 $\langle a \rangle \cap \langle b \rangle = \langle \text{lcm}(a, b) \rangle$ [If R commutative ring] 但注意! 要在能够用 Division Algorithm | 有 gcd, lcm 的环里用.

Principal Ideal: Ideal I : iff $I = aR$ **Prime Ideal**: Ideal $I \neq R$: $ab \in I \Rightarrow a \in I$ or $b \in I$. (OR: If $a \notin I$ and $b \notin I$, then $ab \notin I$)

Maximal Ideal: Ideal $I \neq R$: If J ideal and $I \subseteq J \subseteq R$, then $J = I$ or $J = R$. $\downarrow$ I not radical $\Leftrightarrow \exists a \notin I$ s.t. $a^n \in I$ for some $n \in \mathbb{N}$

Radical|**Radical Ideal**: $I \neq R$: **Radical**: $\sqrt{I} = \{r \in R : r^n \in I \text{ for some } n \in \mathbb{N}\}$ **Radical Ideal**: If $I = \sqrt{I}$ $I \subseteq \sqrt{I}$ always.

Properties: Relationship between these kinds of ideals:

- R commutative ring, I ideal. **I prime $\Leftrightarrow R/I$ is an ID.** **I maximal $\Leftrightarrow R/I$ is a field**
- R ring, $I_1, I_2, \dots, I_n \triangleleft R$, and $I_1 \cap I_2 \cap \dots \cap I_n = I \triangleleft R \Rightarrow R \cong \text{subring of } R/I_1 \oplus R/I_2 \oplus \dots \oplus R/I_n$. | Moreover, $R/I \cong R/I_1 \oplus R/I_2 \oplus \dots \oplus R/I_n$ if $I_i + I_j = R$ for $i \neq j$
- R ring, I ideal, J ideal of $R/I, T = \{r \in R : r + I \in J\}$ (reps of J). Then $^1 T$: **Ideal of R** $^2 I \subseteq T$ $^3 J = \{t + I : t \in T\}$ $^4 (R/I)/J \cong R/T$
If $J = \langle j_1 + I, \dots, j_n + I \rangle$ then $T = \langle j_1, \dots, j_n \rangle + I = \langle j_1, \dots, j_n, I \rangle$ Moreover, if $I = \langle i_1, \dots, i_m \rangle$, then $T = \langle j_1, \dots, j_n, i_1, \dots, i_m \rangle$
- R commutative ring, I ideal. **I radical $\Leftrightarrow R/I$ has no nonzero nilpotent elements** except 0 (nilpotent: $r^n = 0$)
- If R (commutative with unity), then **any ideal $I \neq R$ is contained in a maximal ideal.** If r is not a unit, then $1 \notin \langle r \rangle = rR$

3 PID | UFD | Euclidean Domain (ED $\Rightarrow$ PID $\Rightarrow$ UFD $\Rightarrow$ ID)

Irreducible: $0 \neq a \in R \setminus R^\times$ is **Irreducible**: If $a = bc$, then $b \in R^\times$ or $c \in R^\times$.

Prime: $0 \neq a \in R \setminus R^\times$ is **Prime**: If $a \mid bc$, then $a \mid b$ or $a \mid c$. (If R commutative ring, then prime $\Rightarrow$ irreducible)

- F field, $0 \neq g \in F[x]$ non-constant: $^1 F[x]/\langle g \rangle$ is a field $\Leftrightarrow$ $^2 g$ is irreducible in $F[x]$ $\Leftrightarrow$ $^3 F[x]/\langle g \rangle$ is a domain
- F field, $0 \neq g \in F[x]$: **g is irreducible $\Leftrightarrow \langle g \rangle$ is prime ideal**
- If R is ID: **Prime $\Rightarrow$ Irreducible** If $0 \neq a \in R$, then: **a prime $\Leftrightarrow aR$ prime ideal**
- If R is PID, $a \neq 0$: $^1 aR$ is prime ideal $\Leftrightarrow$ $^2 a$ is prime $\Leftrightarrow$ $^3 aR$ is maximal ideal $\Leftrightarrow$ $^4 a$ is irreducible

PID: R commutative ring is **principal ideal domain** if: $\forall I \triangleleft R, I = aR$ for some $a \in R$. **Theorem**: $\mathbb{Z}, F, F[x]$ is PID.

ED: R is ID. R is **Euclidean Domain** if: (**Euclidean func**) $\exists v: R \setminus \{0\} \rightarrow \mathbb{N}_0$ s.t. $\forall a, b \in R, b \neq 0, \exists q, r \in R$ s.t. $a = bq + r$ and $r = 0$ or $v(r) < v(b)$.

Common ED: $\mathbb{Z}, F[x], \mathbb{Z}[i]$ Gaussian Integers are ED, with $v(a) = |a|, v(f) = \deg(f), v(a + bi) = a^2 + b^2$ respectively.

Euclidean Algorithm: If R ED, $a, b \in R \setminus \{0\}$, then $\text{gcd}(a, b) = xa + yb$ for some $x, y \in R$. (extended of Bezout's Identity)

UFD (Def 1): R ID is **unique factorization domain** if: **UFD 1**: $\forall a \in R \setminus \{0\}, \exists$ irreducibles $p_1, \dots, p_n \in R$ s.t. $a = p_1 p_2 \dots p_n$

UFD 2: Any other irreducibles $q_1, \dots, q_m \in R$ s.t. $a = q_1 q_2 \dots q_m$, then $n = m$ and $p_i = u_i q_i$ for some $u_i \in R^\times$ after reordering.

Ascending Chain Condition (ACC): A sequence of ideals $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$ satisfies ACC if: $\exists n \in \mathbb{N}$ s.t. $I_n = I_{n+1} = I_{n+2} = \dots$

UFD (Def 2): R ID is UFD if: **UFD 1**: Every sequence of **Principal Ideals** satisfies Ascending Chain Condition (ACC)

UFD 2: If a irreducible, then a is prime ($\Leftrightarrow aR$ is a prime ideal.) **Causs's Theorem**: If R is UFD, then $R[x]$ is UFD.

- Bezout's Identity**: R is PID, then $\forall a, b \in R, \exists x, y \in R$ s.t. $ax + by = \text{gcd}(a, b)$. $\ln \mathbb{F}[\dots, x_n]: \text{lcm}(f, g) = \frac{fg}{\text{gcd}(f, g)}, \langle f, g \rangle = \langle \text{gcd}(f, g) \rangle, \langle f \rangle \cap \langle g \rangle = \langle \text{lcm}(f, g) \rangle$
- R UFD: **a prime $\Leftrightarrow a$ irreducible** **R PID: Every sequence of ideals satisfies ACC.**

4 Localisation

Field of Fractions (F_R): R is ID, $S = R \setminus \{0\}$. Def **equivalence relation** $\sim$ on $R \times S$ by: $(a, b) \sim (c, d) \Leftrightarrow ad = bc$. (Like $\frac{a}{b} = \frac{c}{d}$)

Then the **field of fractions** F_R of R : $F_R = (R \times S) / \sim = \{\frac{a}{b} : a \in R, b \in S\}$ where $\frac{a}{b} := \{(a', b') : (a', b') \sim (a, b)\}$

Operations: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}, \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$, Identity: $0_{F_R} = \frac{0}{1}, 1_{F_R} = \frac{1}{1}$ Inverse: $-\frac{a}{b} = \frac{-a}{b}, (\frac{a}{b})^{-1} = \frac{b}{a}$ if $a \neq 0$.

Remark: $\tau: R \rightarrow F_R$ by $\tau(r) = \frac{r}{1}$ is a 1-1 ring homomorphism. ps: Isomorphism to $\{\frac{r}{1} : r \in R\} \subseteq F_R$. **F_R is a field.**

Multiplicative Set: R commutative ring. $X \subseteq R$ is a **multiplicative set** if: **1**: $1_R \in X$ **2**: 乘法封闭 (option ***3**: $0_R \notin X$)

Ring of Fractions ($X^{-1}R$): R 交换环, $X \subseteq R$ multiplicative set. Def **e.r.** $\sim$ on $R \times X$ by: $(a, x) \sim (b, y) \Leftrightarrow \exists s \in X$ s.t. $s(ay - bx) = 0$.

Then the **ring of fractions** $X^{-1}R$: $X^{-1}R = (R \times X) / \sim = \{\frac{a}{x} : a \in R, x \in X\}$ where $\frac{a}{x} := \{(a', x') : (a', x') \sim (a, x)\}$

Operations: $\frac{a}{x} + \frac{b}{y} = \frac{ay+bx}{xy}, \frac{a}{x} \cdot \frac{b}{y} = \frac{ab}{xy}$, Identity: $0_{X^{-1}R} = \frac{0}{1}, 1_{X^{-1}R} = \frac{1}{1}$ Inverse: $-\frac{a}{x} = \frac{-a}{x}$ If $a \in X$, then $(\frac{a}{x})^{-1} = \frac{x}{a}$.

Remark: $\varphi: R \rightarrow X^{-1}R$ by $\tau(r) = \frac{r}{1}$ is a ring homomorphism. ps: Isomorphism to $\{\frac{r}{1} : r \in R\} \subseteq X^{-1}R$. **$X^{-1}R$ is a ring.**

Localisation: P prime ideal of A . $\Rightarrow$ $^1 S = A \setminus P$ is a **multiplicative set**. 2 Localisation of A at P : $A_P = S^{-1}A$

Theorem: A commutative ring, then: $A_P = S^{-1}A$ has **only one maximal ideal**, which is $S^{-1}P$. ps: $S = A \setminus P$

5 Nullstellensatz, Variety & Vanishing Ideal

5.1 Nil and Jacobson Radical Ideals, Local Ring

Nil Ideal: I ideal of R is a <i>nil ideal</i> if: $\forall x \in I, \exists n \in \mathbb{N} \text{ s.t. } x^n = 0$. i.e. every element of I is nilpotent.
Nil Radical ($N(R) = \sqrt{0_R}$): $N(R) :=$ the largest nil ideal in $R \Leftrightarrow N(R) :=$ sum of all nil ideals in $R \Leftrightarrow N(R) = \sqrt{0_R}$ Note: sum of any two nil ideals is also a nil ideal. Note: $R/N(R)$ has no non-zero nilpotent elements.
Quasi-invertible: R ring, $a \in R$ is <i>quasi-invertible</i> if: $\exists b \in R \text{ s.t. } (1-a)(1-b) = 1 \Leftrightarrow ab = a + b$. (如果 a quasi-invertible, 那么 b 唯一)
Jacobson Radical Ideal: I ideal of R is a <i>Jacobson radical ideal</i> if: $\forall i \in I, i$ is <i>quasi-invertible</i> in R .
Jacobson Radical $J(R)$: $J(R) :=$ the largest JRI in $R \Leftrightarrow J(R) :=$ sum of all JRI in $R \Leftrightarrow J(R) = \bigcap_{(\text{left})} \text{maximal ideals of } R$. Note: sum of any two JRI is also a JRI. Note: Nil ideal $\Rightarrow$ Jacobson Radical Ideal (JRI). (通过等比数列证明)
Local Ring: R is a <i>local ring</i> if: R has unique maximal ideal $\Leftrightarrow R/J(R)$ is a field A_P is a local ring . (A commutative ring, P prime ideal of A)

Remark: $R/(JRI)$ or $R/J(R)$ has no non-zero nilpotent elements $(1-a)(1-b) = 1 \Leftrightarrow 1-a$ is invertible ($b = 1 - (1-a)^{-1}$)

5.2 Variety & Vanishing Ideal

Variety: Let k be a field, $f_1, \dots, f_m \in k[x_1, \dots, x_n]$. Then: Variety: $V(f_1, \dots, f_m) := \{(a_1, \dots, a_n) \in k^n : f_i(a_1, \dots, a_n) = 0, \forall i\}$
Vanishing Ideal: Let k be a field, $X \subseteq k^n$. Then: Vanishing Ideal: $\mathcal{I}(X) := \{f \in k[x_1, \dots, x_n] : f(a_1, \dots, a_n) = 0, \forall (a_1, \dots, a_n) \in X\}$

Useful Tools: If $X = X_1 \cup X_2 \cup \dots \cup X_m$, then $\mathcal{I}(X) = \mathcal{I}(X_1) \cap \mathcal{I}(X_2) \cap \dots \cap \mathcal{I}(X_m)$ · If $I_1 \subseteq I_2$ ideals, then $V(I_2) \subseteq V(I_1)$

5.3 Nullstellensatz

Weak Nullstellensatz: $\mathbb{F}$ field, and $I \triangleleft \mathbb{F}[x_1, \dots, x_n]$ ideal. Then: $(\Rightarrow)$ If $I = \langle x_1 - a_1, \dots, x_n - a_n \rangle, a_i \in \mathbb{F}$, then I is maximal ideal.

Weak Nullstellensatz $(\Leftarrow)$ If I is *maximal ideal* and $\mathbb{F}$ is algebraically closed, then $I = \langle x_1 - a_1, \dots, x_n - a_n \rangle$ for some $a_i \in \mathbb{F}$.

Fact I: If $I = \langle x_1 - a_1, \dots, x_n - a_n \rangle$, then $[f(a_1, \dots, a_n) = 0 \Leftrightarrow f \in I]$

Fact II: k field, $(\Rightarrow)$ I maximal ideal then **$V(I)$ is a single point.** (i.e. $V(I) = \{(a_1, \dots, a_n)\}$) $(\Leftarrow)$ if k algebraically closed. (ps: 最好证明一下)

Reformation of Weak Nullstellensatz: k algebraically closed field, J ideals of $k[x_1, \dots, x_n]$. Then:

$^1 V(J) \neq \emptyset \Leftrightarrow J \neq k[x_1, \dots, x_n]$ $^2 f \in \mathcal{I}(V(J)) \Leftrightarrow f^n \in J$ for some $n \in \mathbb{N}$ i.e. $\sqrt{J} = \mathcal{I}(V(J))$ It's Strong Nullstellensatz!

Reformation of Hilbert's Nullstellensatz: k algebraically closed field, I ideal of $k[x_1, \dots, x_n]$. Then:

$^1 \sqrt{I} = \bigcap (\text{maximal ideals} \supseteq I) \Leftrightarrow ^2 J(R/I) = \bigcap \text{maximal ideals in } R/I = \text{nil radical of } R/I = \sqrt{0_{R/I}}$

2nd Def of Radical: R commutative ring, I ideal of R . Then: $\sqrt{I} = \bigcap (\text{prime ideals} \supseteq I)$

Corollary: If R commutative ring, then **Nil radical** $\sqrt{0_R} = N(R) = \bigcap \text{prime ideals of } R =:$ **Prime Radical.** If R 非交换, 不一定.

6 Gröbner Bases

Monomials: A <i>monomial</i> is a product of elements from $\{x_1, x_2, \dots, x_n\}$. i.e. $f = x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}$ (if 交换) $\deg(f) = \sum_i \alpha_i$
Lex Order (Lexicographic Order): 1 定义变量顺序; 2 看第一个不同的变量 e.g. $x > y > z$, then $x > yxx; xy > yx$ (If NOT 交换)
Deglex Order: For f, g monomials with variables $x_1, x_2, \dots, x_n$: 1 Compare degree; 2 If equal, define ordering on variables (e.g. $x_1 > x_2 > \dots > x_n$) and compare lexicographically. If NOT 交换: $xyx > yxx$
multideg LC LM LT & Ov: 分别为: 排序后的最大的一项的: 指数向量 系数 单项式 (coeff=1) 整项 (完整的) Overlap (Ov): f, g poly, $Ov(f, g) = \gcd(LM(f), LM(g))$
S-polynomial: For polynomials f, g : $S(f, g) = \frac{\text{lcm}(LM(f), LM(g))}{LT(f)} f - \frac{\text{lcm}(LM(f), LM(g))}{LT(g)} g = \text{or } \frac{LT(g)}{Ov(f, g)} \cdot f - \frac{LT(f)}{Ov(f, g)} \cdot g$ 效果: f, g 的 LM 减小
Gröbner Basis: I ideal of $k[x_1, \dots, x_n], k$ field. A subset $G \subseteq I$ is a <i>Gröbner basis</i> of I with respect to a <i>monomial order</i> (e.g. Lex, Deglex) if: $\forall f \in I, f \neq 0, \exists g \in G \text{ s.t. } LM(g) \mid LM(f)$. $(\Leftrightarrow)$ For any w_i monomials, $\alpha \in k$, s.t. if $LM(g) \nmid w_i$ for all $g \in G$, then $\sum_i \alpha_i w_i \notin I$.

Algorithm to Find Gröbner Basis: For ideal $I = \langle f_1, f_2, \dots, f_m \rangle \triangleleft k[x_1, \dots, x_n], k$ field. e.g. Buchberger's Algorithm

- Let $G = \{f_1, f_2, \dots, f_m\}$
- For each $g_i \in G$, write $g_i = LC(g_i) \cdot LM(g_i) + h_i$
- For each pair (g_i, g_j) in G , calculate $S(g_i, g_j)$ only $i > j$ or $i < j$ to avoid repetition **Remark:** If $Ov(g_i, g_j) = 1$, then $S(g_i, g_j) = 0$, skip it.
- Reduce each $S(g_i, g_j)$: If $LM(g_k)$ divides any term of $S(g_i, g_j)$, substitute $LM(g_k)$ by $-h_k$; until no $LM(g_k)$ divides any term.
- If $S(g_i, g_j) \neq 0$ after reduction, add it's leading term to G and go to step 2. If all $S(g_i, g_j) = 0$, stop and output G .

Bröbner Basis Application: k field, $I \triangleleft k[x_1, \dots, x_n], G$ Gröbner basis of I . L leading terms of G . W set of monomials not divisible by any element of L . (i.e. don't contain any monomials in L as a subword) Then:

- $\{w + I : w \in W\}$ is a **basis of $k[x_1, \dots, x_n]/I$ as a vector space over k .**
- For each $g_i \in G$, write $g_i = LC(g_i) \cdot LM(g_i) + h_i$, **If $r \in k[x_1, \dots, x_n]$ can be reduced to 0 by $g_i \rightarrow -h_i$, then $r \in I$.**

7 Appendix

Zorn's Lemma: If a partially ordered set (i.e. $\exists \sim$) P s.t. $^1 \forall$ chain in P has upper bound. $^2 P \neq \emptyset \Rightarrow P$ has at least one maximal element. (i.e. $\exists m \in P \text{ s.t. } \forall p \in P, \text{ if } m \leq p, \text{ then } m = p$)

2nd Def of Prime Ideal: R commutative ring, I ideal of R . Then: **I prime ideal** $\Leftrightarrow [J, K \text{ ideals s.t. } JK \subseteq I \Rightarrow J \subseteq I \text{ or } K \subseteq I] \Leftrightarrow [J \subseteq I, K \subseteq I \Rightarrow JK \subseteq I]$